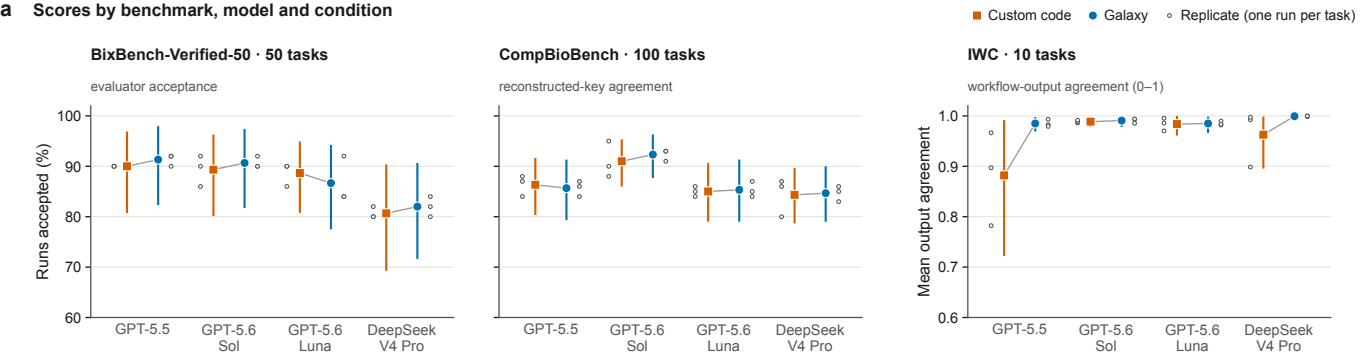
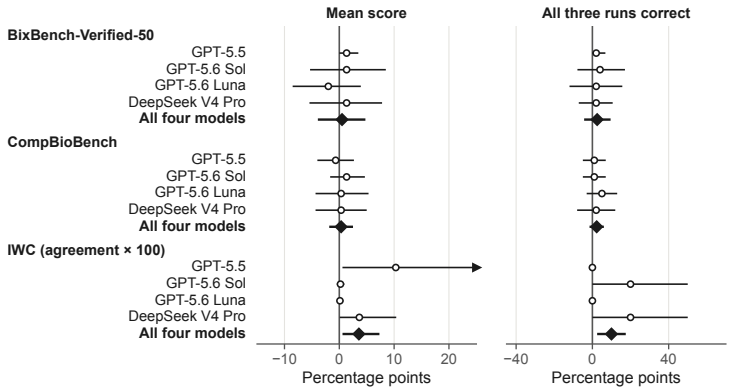
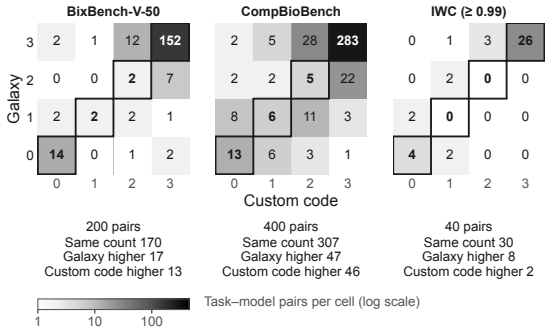
a Scores by benchmark, model and condition



b Paired differences: Galaxy minus custom code

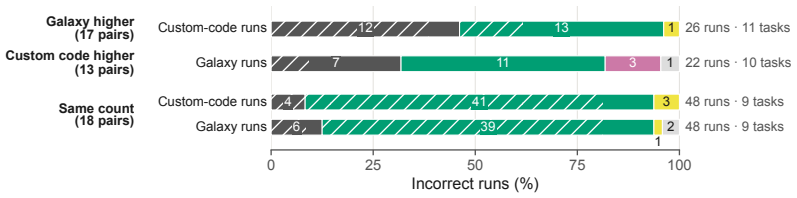


c Correct runs of three, same task and model



d Why the conditions disagree (BixBench-Verified-50)

Incorrect runs by primary cause (AI-assisted audit), for task–model pairs whose conditions differ or agree
An independent second rater agreed on the cause for 35 of 45 sampled runs (κ = 0.56)



Traced case: bix-45-q1 (relative composition variability)

	Custom code	Galaxy
GPT-5.5	○ ○ ○	○ ○ ○
GPT-5.6 Sol	● ● ●	○ ○ ○
GPT-5.6 Luna	● ○ ●	○ ○ ○
DeepSeek V4 Pro	● ● ●	○ ○ ○
	8 of 12 correct	0 of 12 correct

Successful custom-code runs matched the PhyKIT version to the date of the input archive. The Galaxy PhyKIT wrapper, like current PhyKIT, returned a different value, and the wrapper version was not shown to the agent; the reference encodes the older value.

● correct ○ incorrect (one run each)